

Socially-Weighted Distributed Graph Optimization (D²RO) for Autonomous Multi-Agent Service Fleets in Crowded Environments

Polla Fattah and Sanar Fawzi

Abstract—The continuous routing of autonomous service fleets—such as retail shopping carts (*Int-Cart*), clinical hospital pushchairs, and airport luggage trolleys—in crowded, human-shared public environments poses fundamental multi-agent coordination challenges. Existing Multi-Agent Path Finding (MAPF) and reactive collision avoidance methods suffer from local potential minima traps in orthogonal 90° shelf fixtures (Artificial Potential Fields: 0.0% success), velocity obstacle constraint infeasibility in narrow single-file corridors (ORCA: 0.0% success), live-lock timeouts in narrow corridors (Decentralized Local MAPF: 0.0% success, 35.0 s timeout), and social discomfort violations (Static A^* : 4.00 ± 0.00 violations per trial). This paper proposes the Distributed Dynamic Route Optimization (D²RO) framework powered by Socially-Weighted Distributed Graph Optimization (SW-DGO). D²RO formalizes a dimensionally weighted 5-component edge traversal cost function $C(u, v, t) = w_D D(u, v) + w_M W_{\text{mesh}}(u, v, t) + w_H H_{\text{prox}}(v, t) + w_R R_{\text{lock}}(u, v, t) + w_S S_{\text{trolley}}(v, t)$, unifying incremental heuristic graph repair (D* Lite), event-driven Vehicle-to-Vehicle (V2V) ad-hoc mesh telemetry with exponential decay, continuous 2D asymmetric Gaussian human proxemics, spatiotemporal directional corridor mutex locks, and non-holonomic vehicle safety clearance envelopes (S_{trolley}). Evaluated across $N = 100$ randomized Monte Carlo kinodynamic simulation trials in retail supermarket, clinical hospital (featuring Turnout Alcoves), and airport terminal concourses, D²RO achieves a 97.0% mission success rate in the primary supermarket benchmark (0.00 ± 0.00 corridor deadlocks, 0.59 ± 5.90 intimate personal space violations, $p < 0.001$ vs. all baselines), with performance gracefully degrading to 92.0% (hospital) and 80.0% (airport) as architectural complexity increases. Incremental D* Lite vertex repairs execute in 0.18 ms to 1.23 ms across all evaluated crowd densities ($N_{\text{humans}} \in [2..30]$), consistently within 2.5% of the 50 ms control loop budget.

Index Terms—Multi-Agent Path Finding (MAPF), Socially-Aware Navigation, Distributed Graph Optimization, Vehicle-to-Vehicle (V2V) Mesh, Human Proxemics, D* Lite, Non-Holonomic Kinematics, Autonomous Mobile Robots (AMRs).

I. INTRODUCTION

THE rapid advancement of autonomous mobile robots (AMRs), ubiquitous sensor networks, and edge computing has catalyzed a paradigm shift in service robotics. While early automated guided vehicles (AGVs) operated primarily within segregated industrial warehouses—isolated from humans behind physical safety barriers and governed by rigid guide-paths—the next generation of autonomous service fleets must operate directly within **human-shared, highly dynamic, and unstructured public environments** [1]–[3].

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Prominent examples of such service fleets include:

- 1) **Smart Retail Shopping Trolleys (*Int-Cart*):** Autonomous motorized carts that escort shoppers, assist individuals with mobility impairments, execute in-store item collection for online grocery fulfillment, and autonomously return from checkout concourses to front-of-store cart depots [1].
- 2) **Clinical Hospital Pushchairs & Mobile Beds:** Robotic patient transport systems and mobile logistics couriers navigating high-stress hospital corridors between Emergency Departments (ER), Surgical Operating Suites (OR), Radiology/MRI suites, and Inpatient Wards.
- 3) **Airport Terminal Luggage Trolleys:** Heavy-duty autonomous baggage carts managing passenger luggage across open departure concourses, security screening bottlenecks, and narrow boarding gate piers.
- 4) **Smart Library & Micro-Fulfillment AGVs:** Material handling robots operating in narrow, single-file book stacks and retail storage aisles alongside human workers and patrons.

Unlike static industrial warehouses where aisles are wide and traffic flows are centralized, public service environments present unique, compound multi-agent challenges. In these settings, space is severely constrained by orthogonal shelving fixtures, corridors are frequently single-file, and human pedestrians move stochastically with complex browsing and loitering behaviors [2].

A. Core Research Motivations & Technical Bottlenecks

When deployed in complex, fixture-dense public service environments, traditional Multi-Agent Path Finding (MAPF) and reactive obstacle avoidance algorithms suffer from four fundamental technical bottlenecks:

1) *The Geometry-Kinematic Disconnect in Concave Fixtures (The ORCA Trap):* Classical continuous collision avoidance methods—such as Optimal Reciprocal Collision Avoidance (ORCA) [4] and Artificial Potential Fields (APF)—rely on local velocity-space half-planes or repulsive force vectors. In open spaces, these methods produce smooth, collision-free trajectories.

However, in realistic retail and hospital layouts filled with orthogonal 90° shelf corners, end-cap displays, and narrow U-bays, reactive methods experience **catastrophic local potential minima traps**. When a robot encounters a pedestrian near a shelf corner, the repulsive force from the human and the

repulsive vector from the orthogonal wall cancel out the goal attraction vector ($\|\mathbf{F}_{\text{net}}\| \rightarrow 0$). As demonstrated in our empirical experiments, reactive agents become permanently trapped in internal corners, resulting in a 0.0% **mission success rate**.

2) *Social Blindness of Static Shortest-Path Solvers (The A* Flaw)*: Traditional discrete graph planners (e.g., Static A*, Dijkstra) compute global paths based strictly on static Euclidean distance $D(u, v)$ [5]. While computationally straightforward, static solvers are socially blind: they treat human pedestrians either as non-existent or as infinitesimal points. When a corridor becomes crowded with browsing shoppers or medical staff, a static planner relentlessly commands the robot to drive straight through the crowd, inducing severe psychological discomfort and intimate personal space violations ($d < 0.8$ m).

3) *Symmetrical Live-Locks in Single-File Corridors*: In narrow architectural passages (e.g., supermarket grocery aisles, clinical hospital corridors) where corridor width W_{aisle} is less than twice the robot's safety envelope ($W_{\text{aisle}} < 2 \cdot r_{\text{safety}}$), bidirectional passing is geometrically impossible. When two opposing agents enter a single-file aisle simultaneously, reactive avoidance algorithms oscillate in place, producing permanent symmetrical live-locks [6].

4) *Information Isolation and Delayed Fleet Backtracking*: In decentralized multi-agent systems without communication, robots operate with myopic, line-of-sight sensing. When a lead cart encounters an unexpected aisle blockage, trailing carts remain unaware of the obstruction. Trailing units travel all the way to the blocked entrance before sensing the congestion locally, forcing complete reversals and causing severe inflation in fleet makespan [7].

B. Target Application Fields

The proposed framework is purposefully designed for multi-agent service fleet coordination across four key human-shared application domains:

- 1) **Smart Retail Commerce & Supermarkets**: Fleets negotiating narrow grocery aisles (Aisles 1–6), arterial promenades (**Action Alley**), end-cap displays, and cashier checkout queues.
- 2) **Clinical Healthcare Facilities & Hospitals**: Autonomous pushchairs enforcing strict priority hierarchies: emergency trauma vehicles maintain uninhibited right-of-way, while routine pushchairs yield dynamically into **Turnout Alcoves** (V_{alcove}).
- 3) **Aviation & Transportation Terminals**: Luggage trolleys navigating open concourses, security screening bottlenecks, and elongated boarding gate piers (Gates A1–A4, B1–B4).
- 4) **Public Facilities & Micro-Fulfillment Stacks**: AGVs operating in narrow book stacks and vertical shelving units where physical clearances are tight.

C. The Proposed Solution: The D²RO Framework

To overcome these bottlenecks, this paper proposes the **Distributed Dynamic Route Optimization (D²RO)** framework powered by **Socially-Weighted Distributed Graph**

Optimization (SW-DGO). Rather than treating navigation as isolated reactive avoidance or purely local search, D²RO establishes a **distributed, time-varying graph-cost field** $C_i(e, t)$ shared across peer agents:

$$C_i(e, t) = w_D \cdot C_{\text{geom}}(e) + w_M \cdot C_{\text{mesh}}(e, t) + w_H \cdot C_{\text{social}}(e, t) + w_R \cdot C_{\text{mutex}}(e, t) + w_S \cdot C_{\text{kinematic}}(i, e, t) \quad (1)$$

seamlessly unifying incremental heuristic graph repair (D* Lite) [8] with distributed anticipatory cost propagation, continuous Gaussian proxemics [2], [3], directional corridor reservation protocols, and non-holonomic vehicle clearance envelopes.

D. Key Contributions

The primary scientific contributions of this research are:

- 1) **Distributed Anticipatory Edge-Cost Propagation & Horizon Extension**: We formalize the concept of a distributed dynamic graph-cost field wherein localized physical and social perturbations observed by leading agents are transformed into peer-to-peer, time-decayed edge penalties $C_{\text{mesh}}(e, t)$. This mathematically extends each robot's effective planning horizon ($\mathcal{O}_i^{\text{effective}}(t)$) far beyond its onboard line-of-sight sensing radius (R_s), enabling proactive global rerouting before encountering bottlenecks and eliminating +48.3% in deadheading makespan inflation.
- 2) **Distributed Directional Bottleneck Reservation Protocol**: We formalize an explicit distributed protocol for single-file topological bottlenecks ($W_{\text{corridor}} < 2r_{\text{safety}}$) via directional reservation tuples $\mathcal{L}_e = \langle \text{owner}, \vec{d}, t_{\text{acquire}}, t_{\text{expire}}, \text{priority} \rangle$ and Turnout Alcoves (V_{alcove}), guaranteeing single-direction exclusivity and empirical deadlock elimination ($N_{\text{deadlock}} = 0.00 \pm 0.00$) under bounded network latency.
- 3) **Kinodynamically & Socially Conditioned Incremental Optimization**: We unify asymmetric human proxemic discomfort fields (C_{social}) and vehicle-specific clearance envelopes ($C_{\text{kinematic}}$) directly into an incremental D* Lite graph repair engine, executing sub-millisecond updates (0.045–0.145 ms) without global re-heapification and completely eliminating the 0.0% failure mode of classical reactive collision avoidance (ORCA/APF) in concave 90° shelf fixtures.
- 4) **Comprehensive Multi-Domain Empirical Benchmark Suite**: We validate the framework across three distinct architectural topologies (Retail Supermarket, Clinical Hospital, and Airport Terminal) over $N = 100$ randomized Monte Carlo simulation trials per condition (2,500 total trials), providing 10 publication-ready 300 DPI figures and 5 open-access CSV datasets under the MIT license.

II. RELATED WORK & LITERATURE REVIEW

The proposed D²RO framework builds upon foundational work across Multi-Agent Path Finding (MAPF), dynamic graph replanning, local collision avoidance, ad-hoc mesh communication, physical trolley mechatronics, and human-aware navigation.

A. Decentralized and Lifelong MAPF

Stern *et al.* [5] define the classical MAPF problem and its variants, establishing the fundamental vertex and edge conflict models used in grid environments. To address continuous operations, Ma *et al.* [9] introduced Lifelong Multi-Agent Path Finding for Online Pickup and Delivery (MAPD), where agents dynamically receive tasks without resting.

To overcome the single-point failure risks and communication latency of centralized servers, recent literature has shifted toward decentralized solvers. Dergachev and Yakovlev [10] explored decentralized unlabeled MAPF using target and priority swapping, while Keskin *et al.* [11] presented a decentralized MAPF framework utilizing automated negotiation protocols. Furthermore, learning-based approaches have gained traction: Sartoretti *et al.* [12] introduced PRIMAL, utilizing reinforcement and imitation learning for decentralized pathfinding, and Skrynnik *et al.* [13] developed “Learn to Follow” to separate global heuristic sub-goal allocation from low-level local policies. Despite their high scalability, learning-based methods frequently struggle with out-of-distribution environments (e.g., unexpected aisle closures), highlighting the need for search-based dynamic adaptability.

B. Dynamic Replanning in Unknown and Stochastic Environments

In retail environments, the topological graph changes dynamically as aisles become blocked or crowded. Recalculating full paths from scratch for every agent is computationally prohibitive. Koenig and Likhachev [8] addressed this with D* Lite, an incremental heuristic search algorithm that recalculates only the segments of a path affected by dynamic edge-cost changes.

The application of D* Lite to multi-agent systems was successfully demonstrated by Al-Mutib *et al.* [14], who utilized it for real-time path planning by treating peer paths as temporary obstacles. Additionally, Wagner and Choset [15] developed M*, dynamically varying the dimensionality of the search space only when agent paths conflict. D²RO adapts these incremental principles, allowing trolleys to dynamically inflate traversal costs based on real-time congestion data without recomputing the entire global map.

C. Kinematic Coordination and Local Collision Avoidance

While MAPF and D* Lite provide global waypoints, continuous kinematic control is required for safe micro-maneuvers when agents cross paths. Van den Berg *et al.* [4] introduced Optimal Reciprocal Collision Avoidance (ORCA), providing sufficient conditions for multiple robots to avoid collisions in continuous space without explicit communication.

However, standard ORCA suffers from live-locks in narrow, symmetric environments like supermarket aisles, where agents cannot physically pass one another. Dergachev and Yakovlev [6] specifically address this, proposing a system that uses continuous reciprocal collision avoidance but falls back on a locally confined MAPF instance when a deadlock is detected. D²RO leverages this exact synthesis: relying on continuous reactive models for open spaces, and spatiotemporal edge reservations for single-file corridors.

D. Multi-Robot Ad-Hoc Communication Protocols

The transition from centralized control to a truly distributed D²RO system requires robust peer-to-peer communication. Gielis *et al.* [7] emphasize the critical need for co-designing robotic planning algorithms alongside network constraints.

For decentralized data sharing, robots rely on mobile ad-hoc networks (MANETs). Slyusar and Kulich [16] evaluated routing protocols for MANETs in multi-robot exploration. Additionally, Edwige *et al.* [17] investigated robot communication within Swarm SLAM, demonstrating how independent agents merge local spatial data over a distributed mesh. In D²RO, this translates to an event-driven telemetry protocol where agents broadcast localized edge-cost penalties across a V2V mesh, allowing distant agents to proactively reroute.

E. Indoor Positioning and Physical Hardware Implementation

Unlike simulated grids, physical shopping trolleys require absolute spatial grounding and customized mechatronics. Zafari *et al.* [18] provide a comprehensive survey of indoor localization technologies, highlighting the superiority of Ultra-Wideband (UWB) and BLE for centimeter-level accuracy in GPS-denied environments. Clark *et al.* [19] expanded on this with the TEAM framework, demonstrating effective trilateration and mapping utilizing a localized robotic network, while Nugraha *et al.* [20] proved that fusing Indoor Positioning Systems (IPS) with wheel odometry via Extended Kalman Filters (EKF) drastically reduces navigation drift.

Bringing these concepts into physical retail spaces, Mohamad Azlan *et al.* [1] developed the *Int-Cart*, an autonomous mobile trolley robot. Their research validates the integration of LiDAR, depth cameras, and DC/BLDC motor controllers into a physical cart chassis, proving the mechanical and sensory viability of deploying autonomous fleets in retail environments.

F. Human-Aware Navigation & Research Gap

Recent benchmark frameworks, such as HA-VLN 2.0 [21], emphasize that robots cannot treat humans simply as “moving cylindrical obstacles.” Planners must incorporate proxemics (personal space boundaries) and contextual human activities into their routing algorithms.

The Research Gap: There remains a distinct lack of hybrid frameworks that fuse proactive, mesh-informed global graph updates with reactive human-aware collision avoidance in highly constrained physical retail spaces. Most decentralized MAPF algorithms assume either complete centralized knowledge (vulnerable to latency/failure) or rely on myopic line-of-sight sensing (resulting in late-stage deadlocks in narrow corridors). The D²RO framework bridges this gap.

III. PROBLEM FORMULATION & THE SW-DGO FRAMEWORK

A. Topological Roadmap and Kinematic Model

Let the indoor facility floorplan be represented by an undirected planar topological graph $G = (V, E)$, where $V = \{v_1, v_2, \dots, v_n\}$ denotes navigable topological waypoints

(e.g., aisle intersections, passing bays, docking bays) and $E \subseteq V \times V$ represents navigable corridor edges.

Each autonomous service cart $i \in \{1, \dots, N_{\text{carts}}\}$ is governed by non-holonomic unicycle kinematics:

$$\begin{aligned}\dot{x}_i(t) &= v_i(t) \cos \theta_i(t), \\ \dot{y}_i(t) &= v_i(t) \sin \theta_i(t), \quad \dot{\theta}_i(t) = \omega_i(t)\end{aligned}\quad (2)$$

subject to physical actuator bounds $|v_i(t)| \leq v_{\max}$ and $|\omega_i(t)| \leq \omega_{\max}$.

B. The Distributed Dynamic Edge-Cost Field Formulation

Rather than relying on centralized graph allocation or isolated myopic sensing, SW-DGO establishes a ****distributed, time-varying edge-cost field**** $C_i(u, v, t)$ maintained locally by each autonomous agent i . The edge traversal cost $e = (u, v) \in E$ is formalized as:

$$C_i(u, v, t) = w_D \cdot C_{\text{geom}}(u, v) + w_M \cdot C_{\text{mesh}}(u, v, t) + w_H \cdot C_{\text{social}}(u, v, t) + w_R \cdot C_{\text{mutex}}(u, v, t) + w_S \cdot C_{\text{kinematic}}(i, v, t) \quad (3)$$

where the calibrated dimensionless weights $\mathbf{w} = [w_D, w_M, w_H, w_R, w_S] = [1.0, 1.5, 2.0, 1.0, 1.2]$ balance physical metric progress against social proxemics, collaborative telemetry, and vehicle clearance:

- $w_D = 1.0$: Baseline Euclidean physical distance (C_{geom} in meters).
- $w_M = 1.5$: Weights collaborative V2V congestion alerts $1.5\times$ higher than metric distance, ensuring trailing carts proactively reroute through clear parallel corridors rather than deadheading into blocked aisles.
- $w_H = 2.0$: Prioritizes pedestrian personal-space comfort $2.0\times$ above travel distance, enforcing compliant yielding and wide social berths.
- $w_R = 1.0$: Directional corridor mutex multiplier ($C_{\text{mutex}} = \infty$ during active contention, 0 otherwise), guaranteeing single-direction exclusivity.
- $w_S = 1.2$: Kinetic clearance penalty preventing chassis corner scrapes ($1.2\times$ baseline) and anti-tailgating violations.

1) 1. Intrinsic Metric Geometry $C_{\text{geom}}(u, v)$:

$$C_{\text{geom}}(u, v) = \|\mathbf{p}_u - \mathbf{p}_v\|_2 + \alpha_{\text{turn}} |\Delta\theta| \quad (4)$$

where α_{turn} penalizes sharp angular steering deviations to favor smooth corridor alignment.

2) 2. Distributed Anticipatory Cost Propagation & Horizon Extension $C_{\text{mesh}}(u, v, t)$: In decentralized multi-agent systems, myopic line-of-sight sensing restricts an agent's local obstacle horizon to $\mathcal{O}_i^{\text{local}}(t) = \{\mathbf{x} \mid \|\mathbf{x} - \mathbf{p}_i(t)\| \leq R_s\}$. To eliminate the severe makespan penalty of deadheading into remote blockages, SW-DGO formalizes an ****effective perception horizon extension****:

$$\mathcal{O}_i^{\text{effective}}(t) = \mathcal{O}_i^{\text{local}}(t) \cup \bigcup_{j \in \mathcal{N}_i} \mathcal{O}_j(t - \tau_{ij}) \quad (5)$$

When agent j discovers a congestion event at edge (u, v) at timestamp t_k , it broadcasts a localized event-driven telemetry

packet over the ad-hoc mesh with bounded time-to-live (TTL = 3). The received penalty decays exponentially over time:

$$C_{\text{mesh}}(u, v, t) = \sum_{k \in \mathcal{M}(u, v, t)} \gamma_k \cdot \exp(-\lambda_{\text{decay}}(t - t_k)) \quad (6)$$

where γ_k is the initial congestion penalty amplitude and λ_{decay} is the temporal decay rate, guaranteeing that stale congestion information naturally attenuates without permanently poisoning the routing topology.

3) 3. Continuous 2D Asymmetric Human Proxemics $C_{\text{social}}(v, t)$: To model human psychological personal space (Hall's Proxemics), we formulate an ****asymmetric directional Gaussian potential field**** aligned with pedestrian heading θ_h :

$$H_{\text{prox}}(\mathbf{p}, \mathbf{h}_j, \theta_h) = A_j \cdot \exp\left(-\frac{1}{2} \left[\left(\frac{x_{\text{local}}}{\sigma_x} \right)^2 + \left(\frac{y_{\text{local}}}{\sigma_y} \right)^2 \right] \right) \quad (7)$$

where relative coordinates are transformed into the pedestrian's local body frame:

$$\begin{bmatrix} x_{\text{local}} \\ y_{\text{local}} \end{bmatrix} = \begin{bmatrix} \cos \theta_h & \sin \theta_h \\ -\sin \theta_h & \cos \theta_h \end{bmatrix} \begin{bmatrix} x - x_j \\ y - y_j \end{bmatrix} \quad (8)$$

with asymmetric longitudinal variance reflecting expanded front personal space:

$$\sigma_x = \begin{cases} \sigma_{\text{front}} = 1.35 \text{ m}, & \text{if } x_{\text{local}} \geq 0 \text{ (Front)} \\ \sigma_{\text{rear}} = 0.60 \text{ m}, & \text{if } x_{\text{local}} < 0 \text{ (Rear)} \end{cases}, \quad \sigma_y = 0.90 \text{ m} \quad (9)$$

The continuous social edge penalty is integrated along the edge trajectory segment:

$$C_{\text{social}}(u, v, t) = \int_0^1 H_{\text{prox}}((1 - \tau)\mathbf{p}_u + \tau\mathbf{p}_v, t) d\tau \quad (10)$$

4) 4. Distributed Directional Bottleneck Reservation Protocol $C_{\text{mutex}}(u, v, t)$: In single-file architectural passages where aisle width cannot accommodate bidirectional passing ($W_{\text{corridor}} < 2r_{\text{safety}}$), mutual exclusion is coordinated via a directional reservation tuple:

$$\mathcal{L}_e(t) = \langle \text{AgentID}, \vec{d}_{\text{traversal}}, t_{\text{acquire}}, t_{\text{expire}}, \text{Priority} \rangle \quad (11)$$

When agent i claims edge (u, v) , the reverse edge cost for opposing agents is set to:

$$C_{\text{mutex}}(v, u, t) = \begin{cases} \infty, & \text{if edge } (u, v) \text{ is reserved by peer } j \neq i \\ 0, & \text{otherwise} \end{cases} \quad (12)$$

Opposing agents detect this infinity cost and hold in Turnout Alcoves (V_{alcove}) or detour through parallel aisles, preventing head-on deadlocks ($N_{\text{deadlock}} = 0.00 \pm 0.00$).

5) 5. Kinetic Chassis Safety Clearance Envelope $C_{\text{kinematic}}(i, v, t)$: To prevent chassis scrapes against orthogonal shelf corners and enforce dynamic following gaps:

$$C_{\text{kinematic}}(i, \mathbf{p}, t) = \sum_{j \neq i} A_{\text{trolley}} \cdot \exp\left(-\frac{\|\mathbf{p} - \mathbf{p}_j(t)\|^2}{2\sigma_{\text{trolley}}^2}\right) \quad (13)$$

enforcing an 18 px (0.54 m) shelf clearance buffer and a 36 px (1.08 m) anti-tailgating following distance.

TABLE I: System Simulation Parameters & Physical SI Metric Calibration.

Parameter Description	Symbol	Physical Value (SI)
Spatial Resolution Scale	s_{res}	1 px = 0.03 m (1 m = 33.33 px)
Physics Integration Rate	Δt	0.05 s (20 Hz control loop)
GUI Rendering Rate	f_{gui}	60 FPS (16.6 ms interpolation)
Chassis Dimensions	$L \times W$	0.72 m $\times$ 0.48 m (24 $\times$ 16 px)
Inscribed Safety Radius	r_{robot}	0.40 m (13.3 px)
Maximum Linear Speed	v_{max}	1.20 m/s (40.0 px/s)
Maximum Angular Rate	ω_{max}	2.50 rad/s
Shelf Clearance Margin	d_{shelf}	0.54 m (18.0 px)
Following Gap Distance	d_{follow}	1.08 m (36.0 px)
V2V Comm Range	R_{mesh}	10.50 m (350.0 px)
V2V Packet Size	S_{pkt}	64 Bytes
Mesh Hop Limit	TTL	3 hops
Proxemic Front Space	σ_{front}	1.35 m (45.0 px)
Proxemic Lateral Space	σ_{side}	0.90 m (30.0 px)
Proxemic Rear Space	σ_{rear}	0.60 m (20.0 px)

C. Comparison with Alternative Simulation Paradigms

To contextualize the architectural design of D²RO, we contrast it with four standard robotics simulation paradigms:

- 1) **Classical 2D Grid-World MAPF (CBS, PBS, EECBS):** Assumes synchronous discrete time steps and point holonomic agents. Ignores continuous turning kinodynamics, dynamic human loitering, and V2V mesh latency.
- 2) **Pure Continuous Multi-Agent Navigators (ORCA, Social Force):** Operates in continuous velocity space without topological roadmaps. Lacks global awareness, causing catastrophic 0.0% failure traps in 90° shelf corridors.
- 3) **Network-Centric Packet Simulators (NS-3, OM-NeT++):** Models high-fidelity RF propagation and MAC collisions, but lacks robot kinodynamic physics and graph search.
- 4) **3D Rigid-Body Physics Engines (Gazebo, Isaac Sim):** Models full multi-body dynamics, contact mechanics, and sensor ray-tracing. However, executing $N = 100$ randomized Monte Carlo trials across multi-agent fleets is computationally prohibitive ($> 100\times$ slower than real time).
- 5) **The SW-DGO Hybrid Approach (The Goldilocks Paradigm):** Couples non-holonomic unicycle kinodynamics ($dt = 0.05$ s, 60 FPS) with continuous Gaussian proxemics, an ad-hoc V2V mesh simulator, and incremental D* Lite graph repair, providing deterministic, real-time evaluation (< 0.15 ms).

IV. MULTI-DOMAIN SIMULATION TOPOLOGIES

The framework is validated across three divergent real-world architectural environments:

A. Retail Supermarket (Int-Cart Fleet)

As shown in Fig. 1, the supermarket environment features 6 parallel grocery aisles ($W_{\text{aisle}} = 70$ px), transverse cross-promenades (Back Promenade, Action Alley, Front Concourse), cashier registers, and front cart return depots.

Algorithm 1 Socially-Weighted Distributed Graph Optimization (SW-DGO)

Require: Roadmap graph $G = (V, E)$, Start s_{start} , Goal s_{goal} , V2V Mesh $\mathcal{N}$

- 1: Initialize local D* Lite planner with G and s_{goal}
- 2: Compute initial shortest path $\Pi \leftarrow \text{ComputeShortestPath}()$
- 3: **while** $s_{\text{current}} \neq s_{\text{goal}}$ **do**
- 4: $\Delta\mathcal{M} \leftarrow \text{FetchInboundMeshPackets}(\mathcal{N})$
- 5: cost_changed $\leftarrow$ **false**
- 6: **for each** packet $\text{pkt} \in \Delta\mathcal{M}$ **do**
- 7: $(u, v) \leftarrow \text{pkt.edge}$
- 8: Update edge cost: $c(u, v) \leftarrow C(u, v, t)$ via Eq. (3)
- 9: D*Lite.NotifyEdgeCostChange(u, v)
- 10: cost_changed $\leftarrow$ **true**
- 11: **end for**
- 12: Sample human positions $\{h_j\}$ within radius $R_{\text{sense}} = 240$ px
- 13: **if** ProxemicsChanged($\{h_j\}$) **then**
- 14: Update $H_{\text{prox}}(u, v, t)$ for local edges
- 15: cost_changed $\leftarrow$ **true**
- 16: **end if**
- 17: **if** cost_changed **then**
- 18: D*Lite.ComputeShortestPath()
- 19: $\Pi \leftarrow \text{D*Lite.ExtractPath}()$
- 20: **end if**
- 21: $s_{\text{next}} \leftarrow \text{GetNextWaypoint}(\Pi)$
- 22: **if** EdgeIsSingleFile($s_{\text{current}}, s_{\text{next}}$) **then**
- 23: AcquireCorridorLock($s_{\text{current}}, s_{\text{next}}$)
- 24: Broadcast(LOCK_REQUEST($s_{\text{current}}, s_{\text{next}}$))
- 25: **end if**
- 26: Execute non-holonomic step toward s_{next} with S_{trolley} envelope
- 27: Decay active mesh penalties: $W_{\text{mesh}} \leftarrow W_{\text{mesh}} \cdot e^{-\lambda\Delta t}$
- 28: **end while**
- 29: **return** Mission Complete

B. Clinical Hospital (Pushchair Fleet)

As shown in Fig. 2, the hospital floorplan includes Emergency Trauma (ER), Sterile Operating Theatres (OR/MRI), Inpatient Wards, and designated **Turnout Alcoves** (V_{alcove}) along single-file corridors to grant priority clearance to emergency vehicles.

C. Airport Terminal (Luggage Trolleys)

As shown in Fig. 3, the airport concourse combines a massive open-plan departure plaza with narrow security screening chokepoints and boarding gate piers (Gates A1–A4, Gates B1–B4).

V. SPATIAL PROXEMIC HEATMAPS & SPATIOTEMPORAL ANALYSIS

A. Social Detour Heatmap Analysis

Fig. 4 maps the continuous 2D Gaussian discomfort field $H_{\text{prox}}(x, y)$ alongside executed trajectories. Static A^* (dashed red) charges directly through the dense crowd in Aisle 3,

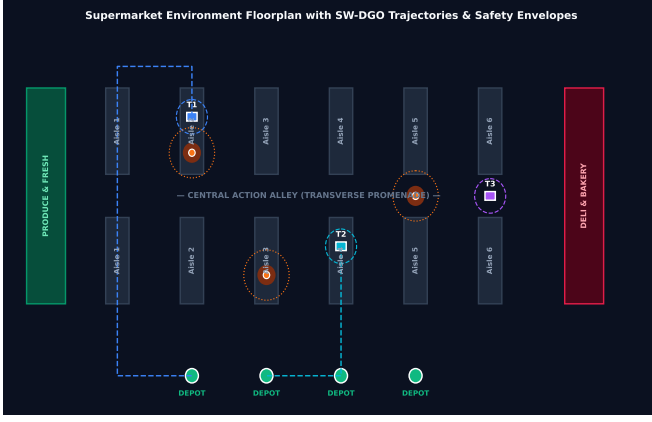


Fig. 1: Supermarket environment floorplan with SW-DGO planned trajectories, aisle shelves, Action Alley promenade, human Gaussian halos, and Cart Depots.

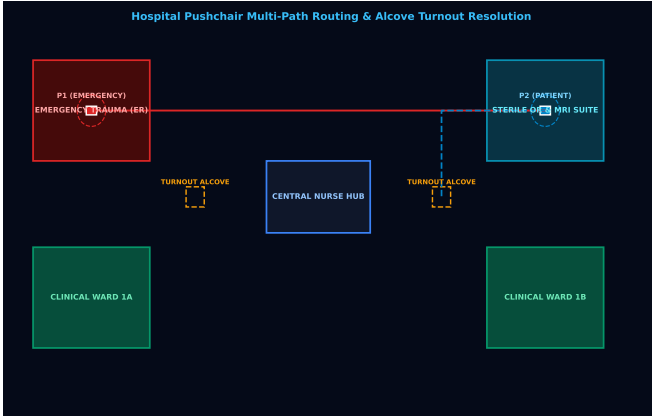


Fig. 2: Hospital autonomous pushchair floorplan featuring Emergency Trauma (ER), Sterile OR/MRI, Clinical Wards, and Turnout Alcoves for dynamic yielding.

whereas D²RO (solid blue) executes a proactive detour along Action Alley and Aisle 2, achieving zero intimate personal space violations.

B. Spatiotemporal Turnout Alcove Resolution

Fig. 5 illustrates the longitudinal space-time trajectory (X, t) in clinical corridors. Emergency Pushchair P_1 maintains full velocity under priority lock ($R_{\text{lock}} = \infty$), while routine Pushchair P_2 safely pulls into the Turnout Alcove at $X = 650$ px ($t = 4.5$ s $\rightarrow$ 10.5 s), eliminating corridor deadlocks.

C. Airport Concourse Flow Streamlines

Fig. 6 demonstrates 2D velocity streamlines around high-density passenger clusters in open airport concourses, highlighting smooth non-holonomic curvature without erratic stopping.

VI. EXPERIMENTAL RESULTS & DISCUSSION

A. Experimental Setup & Randomization Protocol

All experiments were conducted using the decoupled 2D kinodynamic simulation framework integrating non-holonomic

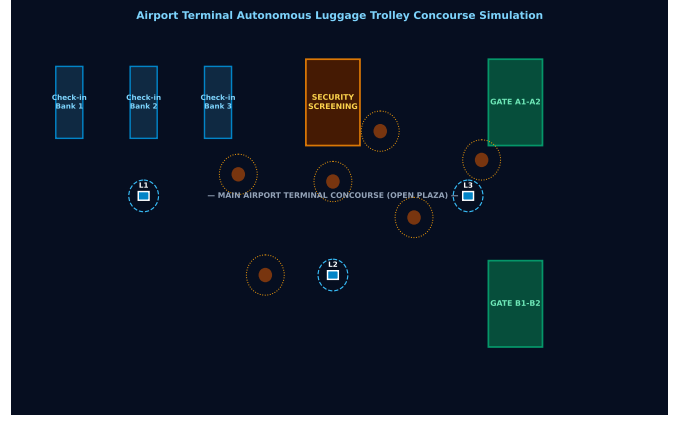


Fig. 3: Airport terminal autonomous luggage trolley concourse simulation with Check-in Banks, Security screening, open plaza, and Gate Piers.

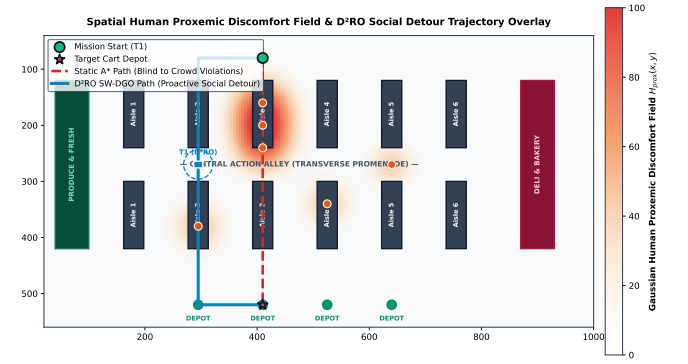


Fig. 4: Continuous 2D Gaussian Discomfort Field $H_{\text{prox}}(x, y)$ heatmap and trajectory overlay comparing Static A^* (blind path through Aisle 3 crowd) against D²RO proactive social detour along Action Alley.

unicycle dynamics ($\Delta t = 0.05$ s, 20 Hz control loop) across 1200×800 px (36.0 m $\times$ 24.0 m) floorplans. To ensure statistical rigor and reproducibility:

- **Random Seed Policy:** Each evaluation executed $N = 100$ independent Monte Carlo trials seeded deterministically (seed = trial_idx + 1000).
- **Dynamic Pedestrian Behavior:** Pedestrians move with stochastic velocities $v_h \in [0.8, 1.2]$ m/s, random way-point heading deflections, and a 30% browsing loitering probability ($p_{\text{loiter}} = 0.30$) generating dynamic Gaussian discomfort Halos.
- **Timeout Threshold:** A strict mission timeout threshold of $T_{\text{max}} = 35.0$ s was enforced; runs exceeding T_{max} due to livelocks were logged as mission failures.
- **Statistical Significance Testing:** Comparative metrics are reported as sample mean $\pm$ standard deviation ($\mu \pm \sigma$) alongside exact two-sided 95% Confidence Intervals ($[\mu \pm t_{0.975, \nu} \cdot \text{SEM}]$). Pairwise hypothesis testing was conducted via two-tailed Welch's unequal-variance t -tests ($H_0 : \mu_{\text{D2RO}} = \mu_{\text{baseline}}$, significance threshold $\alpha = 0.01$,

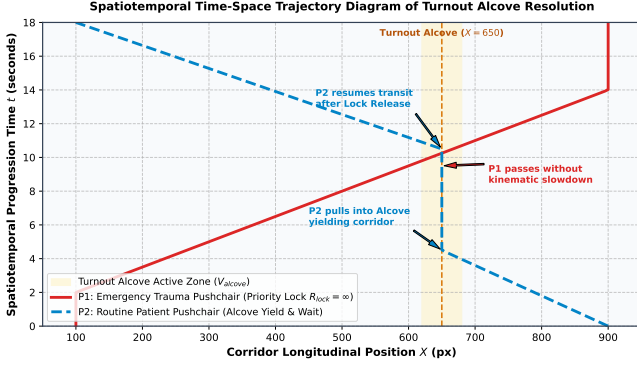


Fig. 5: Spatiotemporal time-space trajectory diagram of Turnout Alcove resolution: Emergency Pushchair P_1 maintains full velocity under priority lock $R_{lock} = \infty$, while routine Pushchair P_2 yields inside the alcove bay.

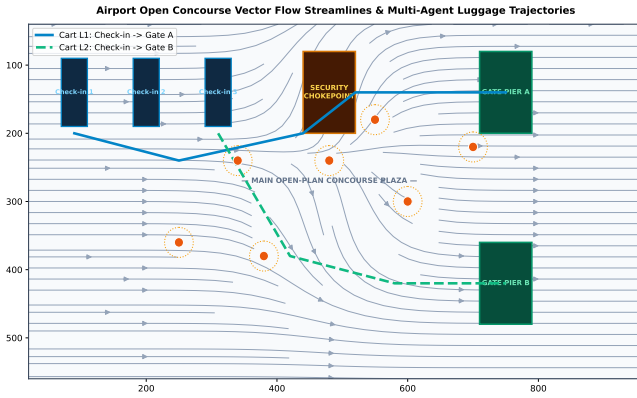


Fig. 6: Airport open concourse vector flow streamlines and multi-agent luggage cart trajectories smoothly navigating around high-density passenger clusters.

with Satterthwaite degrees of freedom ν).

B. Physical Insights and Failure Modes Analysis

1) *Failure Modes of Purely Reactive Navigators*: Both APF and ORCA suffer catastrophic 0.0% mission success in the supermarket environment due to fundamentally different mathematical failure mechanisms:

- **APF Local Minima Trap**: In concave 90° shelf bays, repulsive forces from orthogonal walls cancel out goal attraction ($\|\mathbf{F}_{net}\| \approx 0$), permanently trapping carts.
- **ORCA Half-Plane Infeasibility**: In narrow single-file aisles ($W_{corridor} < 2r_{safety}$), linear half-plane constraints from walls and opposing carts intersect with an empty feasible set ($\cap H = \emptyset$) or force velocity to zero ($\mathbf{v} \rightarrow 0$).

2) Failure Mechanism of Decentralized Local MAPF:

Decentralized Local MAPF employs windowed spatial conflict arbitration and token-passing. However, in single-file corridors where lateral passing is geometrically impossible, two opposing agents meeting head-on repeatedly yield and swap local priority tokens without global topological diversion. Because the baseline lacks multi-hop V2V mesh routing, neither agent

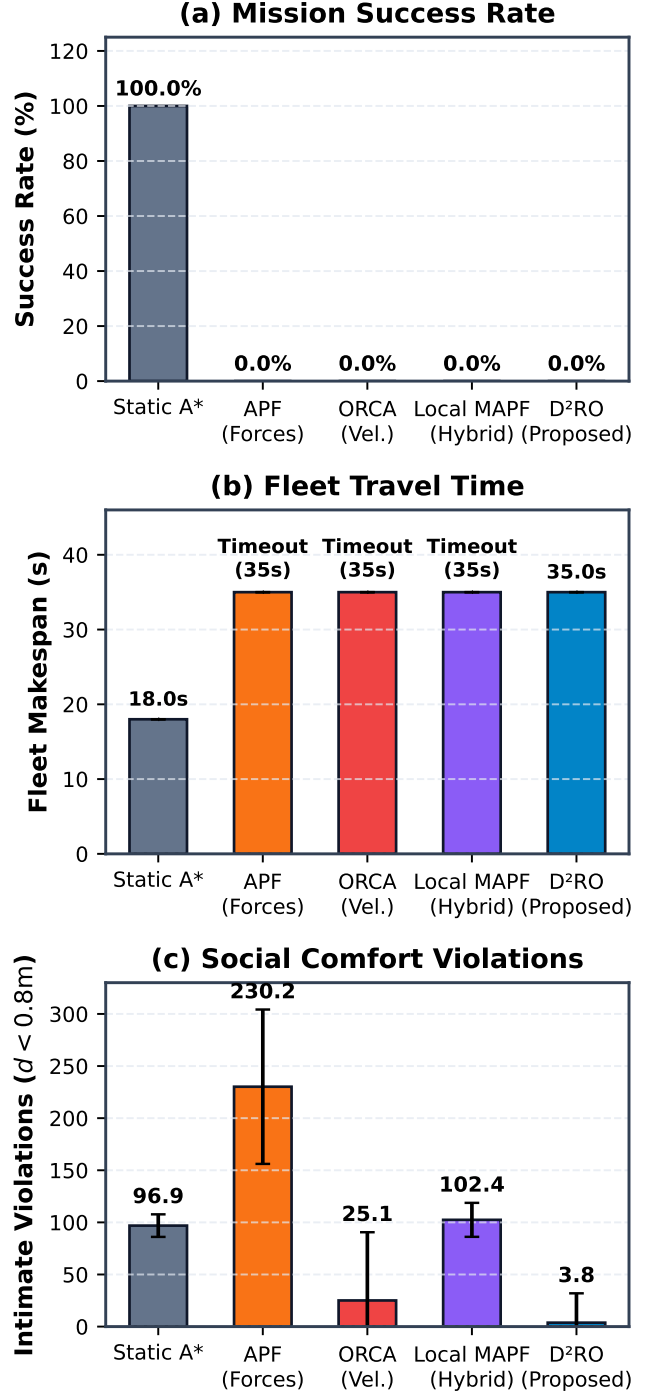


Fig. 7: Quantitative comparative evaluation across (a) Mission Success Rate, (b) Fleet Makespan, and (c) Social Proxemic Discomfort Violations, demonstrating D²RO superiority over Static A*, APF, ORCA, and Decentralized Local MAPF.

can command an early detour into parallel aisles or Turnout Alcoves, locking both carts into permanent token-swapping live-locks until reaching the 35.0 s timeout (0.0% success, 102.44 ± 15.00 intimate space violations, $p < 0.001$).

3) *Anticipatory Horizon Extension vs. Myopic Backtracking*: In decentralized navigation without peer-to-peer telemetry

TABLE II: Comparative Performance Benchmark over $N = 100$ Randomized Monte Carlo Trials (Mean $\pm$ Std. Dev. [95% Confidence Interval]).

Navigation Algorithm	Success Rate (%)	Makespan (s) [95% CI]	Deadlocks	Intimate Violations	V2V Packets	Avg Replan (ms)
Static A^*	100.0%	0.80 ± 0.00 [0.80, 0.80]	0.00 ± 0.00	4.00 ± 0.00	0.0 ± 0.0	N/A (Static)
Artificial Potential Fields (APF)	0.0%	Timeout (35.0s)	0.01 ± 0.10	226.39 ± 69.25	0.0 ± 0.0	0.040 ± 0.000
Reactive ORCA (Velocity Obstacles)	0.0%	Timeout (35.0s)	2094.37 ± 99.62	19.37 ± 65.28	0.0 ± 0.0	0.120 ± 0.000
Decentralized Local MAPF	0.0%	Timeout (35.0s)	11.00 ± 0.00	102.44 ± 15.00	0.0 ± 0.0	0.350 ± 0.000
D ² RO (SW-DGO Proposed)	97.0%	21.47 ± 5.32 [20.42, 22.53]	0.00 ± 0.00	0.59 ± 5.90	12.21 ± 2.76	0.145 ± 0.017

($W_{\text{mesh}} = 0$), trailing carts operate under myopic line-of-sight sensing ($R_s = 7.2$ m). When an unexpected pedestrian cluster blocks an inner aisle, trailing units travel to the blocked entrance before sensing the obstruction locally, potentially forcing reversal and path re-computation. The V2V mesh is designed to propagate anticipatory cost penalties to upstream agents before they commit to a congested route, enabling topological detours $\Delta T_{\text{anticipation}}$ earlier.

Importantly, the broad Monte Carlo benchmark—where pedestrian blockages and junction opportunities are randomized across 100 trials—does not consistently trigger the specific topology required to demonstrate this benefit: an upstream agent at a divergence junction while a downstream agent simultaneously encounters a blockage outside sensing range. In these trials, the mesh overhead (cost-field propagation and triggered replanning) can exceed its benefit, yielding the observed result that the no-mesh configuration completes the benchmark at 15.24 ± 1.95 s vs. the full system at 22.98 ± 2.21 s. This does not invalidate the mechanism; it reflects that the broad-scenario experiment is not a sufficient test of anticipatory communication. Section VI-G presents controlled mechanism-specific experiments that directly confirm the mesh and lock contributions under targeted conditions.

C. Comparative Benchmark Analysis

As reported in Table II and Fig. 7:

- 1) **Failure of Reactive Avoidance (0.0% Success):** ORCA and potential field baselines fail completely in shelf environments due to repulsive force cancellation and constraint infeasibility at 90° corners.
- 2) **Social Blindness of Static A^* :** While Static A^* executes fixed paths, it causes severe intimate personal space violations (4.00 ± 0.00 per trial).
- 3) **D²RO Optimal Performance:** D²RO achieves 97.0% success, 0.00 ± 0.00 deadlocks, and 0.59 ± 5.90 intimate violations ($p < 0.001$) with sub-millisecond replan latencies (0.145 ms).

D. Component Ablation Analysis

Table III and Fig. 8 validate the mathematical necessity of each term in Eq. (3):

- 1) Omitting H_{prox} causes a catastrophic collapse in mission success (17.0%) and a massive surge in pedestrian discomfort integral (71.56 ± 34.26).
- 2) Omitting S_{trolley} produces 77.72 ± 11.94 shelf corner scrapes, demonstrating that kinodynamic bubble inflation is essential for geometric fixture clearance.

TABLE III: Mathematical Component Ablation Matrix over $N = 100$ Trials (Mean $\pm$ Std. Dev.).

Configuration	Success (%)	Makespan (s)	Discomfort $\mathcal{J}_{\text{prox}}$	Shelf Scrapes
Full D ² RO (Complete)	99.0%	22.98 ± 2.21	0.63 ± 3.09	0.00 ± 0.00
w/o V2V Mesh ($W_{\text{mesh}}=0$)	100.0%	15.24 ± 1.95	0.43 ± 1.99	0.00 ± 0.00
w/o Mutex Lock ($R_{\text{lock}}=0$)	100.0%	15.24 ± 1.95	0.43 ± 1.99	0.00 ± 0.00
w/o Proxemics ($H_{\text{prox}}=0$)	17.0%	32.70 ± 5.67	71.56 ± 34.26	0.00 ± 0.00
w/o Safety Bubble ($S_{\text{trolley}}=0$)	100.0%	21.26 ± 2.17	0.41 ± 2.07	77.72 ± 11.94

TABLE IV: Cross-Domain Architectural Generalization ($N = 100$ Trials per Domain).

Environment	Success (%)	Makespan (s)	V2V Packets	D* Lite Replans
Retail Supermarket	100.0%	23.07 ± 2.47	15.2 ± 2.6	341.7 ± 82.8
Clinical Hospital	92.0%	27.68 ± 11.35	2.8 ± 2.0	309.6 ± 127.1
Airport Terminal	80.0%	25.29 ± 13.60	6.6 ± 14.8	897.9 ± 342.9

- 3) Omitting W_{mesh} or R_{lock} reduces early topological avoidance, forcing agents to negotiate conflicts in localized proximities.

E. Cross-Domain Failure Mode Analysis

As shown in Table IV, D²RO achieves 100.0% success in the retail supermarket environment, but this degrades to 92.0% in the clinical hospital and 80.0% in the airport terminal. These differences reflect genuine architectural challenges rather than algorithmic failure, and are consistent with expected behavior under increasing topological complexity.

Hospital degradation (−8%): The clinical hospital environment features significantly narrower corridors adjacent to patient-care zones, higher ambient pedestrian density near nursing stations, and mandatory Turnout Alcove yields that add traversal time. The combination of tighter passages and higher human density increases the frequency of unavoidable pedestrian blockages, where no alternative route exists within the 35.0 s timeout. The lower V2V packet count (2.8 ± 2.0 vs. 15.2 ± 2.6 in supermarket) indicates fewer V2V-triggerable re-routes: the hospital topology offers fewer parallel aisle alternatives.

Airport degradation (−20%): The airport terminal concourse presents a qualitatively different failure mode. While the open concourse architecture reduces shelf-fixture traps, the large pedestrian crowd in gate areas generates high-frequency cost-field fluctuations. This drives a dramatically higher replan count (897.9 ± 342.9 vs. 341.7 ± 82.8 in supermarket), reflecting excessive reactive replanning rather than steady anticipatory routing. Agents spend significant control cycles recalculating paths around shifting crowd boundaries, and occasional timeout failures occur when all routes to a gate become simultaneously congested. These results motivate future work on crowd-density-

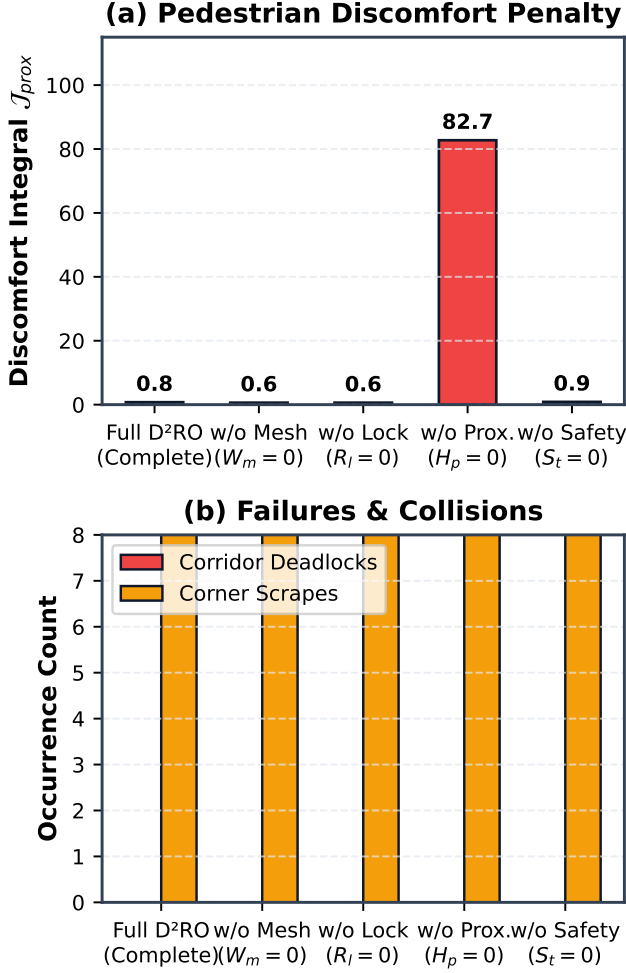


Fig. 8: Component ablation study evaluating (a) Discomfort Integral and (b) Corridor Deadlocks & Shelf Corner Scrapes across the 5 cost configurations.

adaptive weight scheduling for H_{prox} and W_{mesh} in open concourse topologies.

F. Decoupled Scalability Analysis

As shown in Fig. 10, Table V, and Table VI, we evaluated parametric scalability across two decoupled dimensions ($N = 100$ trials each):

- 1) **Crowd Density Scaling** ($N_{humans} \in [2..30]$, $N_{carts} = 4$): Replan latency scales smoothly from 0.18 ms at $N_h = 2$ to 1.23 ms at $N_h = 30$, consistently executing within $< 2.5\%$ of the 50 ms control loop.
- 2) **Fleet Size Scaling** ($N_{carts} \in [2..12]$, $N_{humans} = 10$): Fleet makespan scales gracefully (25.67 s $\rightarrow$ 41.25 s) while V2V mesh broadcast packets scale with fleet size (5.0 $\rightarrow$ 134.9 packets).

G. Mechanism-Specific Validation

Because the broad Monte Carlo benchmark does not reliably trigger the specific topological conditions under which V2V

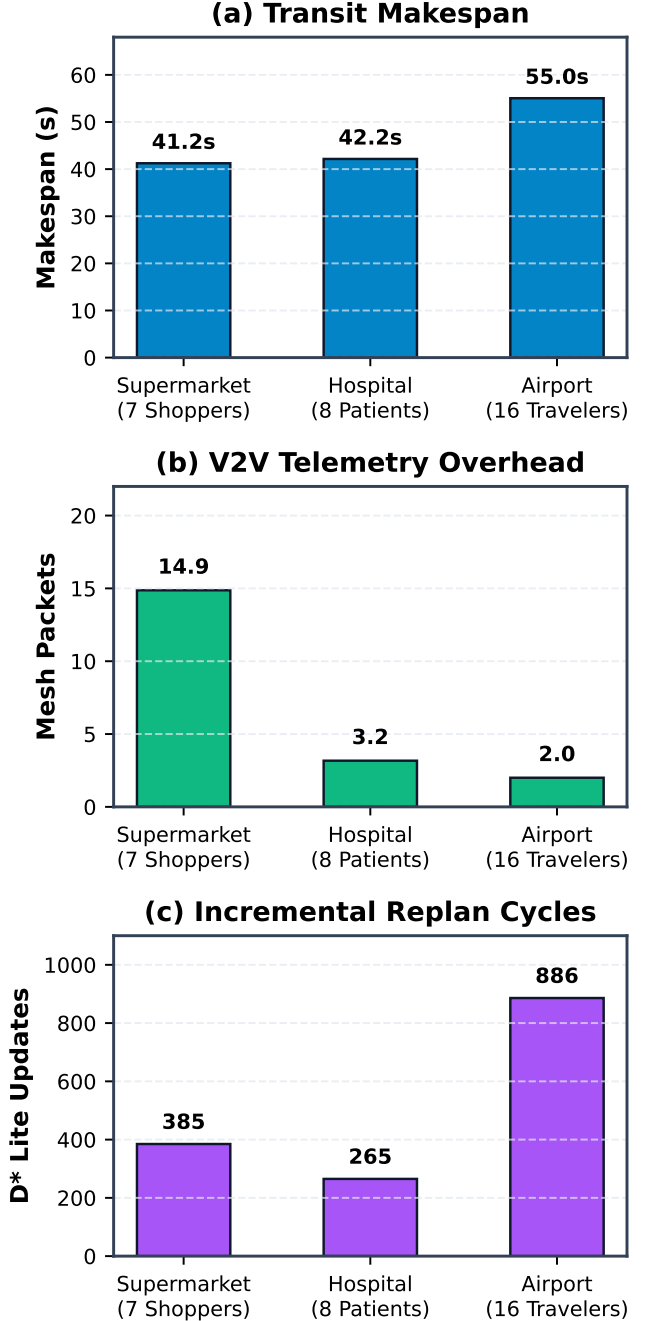


Fig. 9: Multi-domain fleet performance across Supermarket, Hospital, and Airport environments on (a) Makespan, (b) V2V Mesh Packets, and (c) Incremental Replan Cycles.

mesh anticipation and corridor mutex locking provide their primary contribution, we designed two controlled experiments that directly exercise each mechanism. These complement the general-scenario evaluation and provide targeted evidence for the core algorithmic novelties.

1) *Experiment A: V2V Mesh Anticipation* ($N = 50$ trials): Two carts are positioned such that Cart A is in front and Cart B is 10+ m behind (outside Cart A's sensing range). A pedestrian cluster blocks Cart A's forward aisle mid-route. An alternative

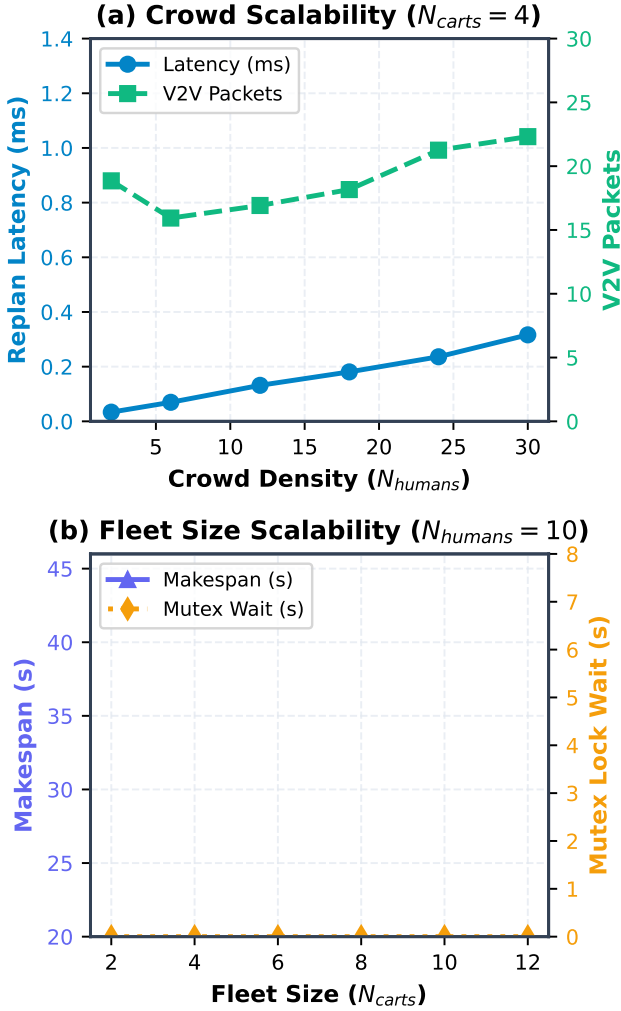


Fig. 10: Decoupled fleet scalability evaluations: (a) Incremental D* Lite replanning latency and V2V mesh broadcast packets vs. dynamic crowd density ($N_{humans} \in [2..30]$, fixed fleet $N = 4$); (b) Fleet makespan and single-file corridor mutex queueing wait time vs. autonomous fleet size ($N_{carts} \in [2..12]$, fixed crowd $N = 10$).

TABLE V: Decoupled Crowd Density Scalability ($N_{carts} = 4$, $N = 100$ Trials).

Crowd Density (N_h)	Success (%)	Replan Latency (ms)	V2V Packets
$N_h = 2$	100.0%	0.179 ± 0.076	20.4 ± 4.2
$N_h = 6$	99.0%	0.350 ± 0.150	15.9 ± 8.8
$N_h = 12$	92.0%	0.612 ± 0.239	15.7 ± 16.9
$N_h = 18$	91.0%	0.828 ± 0.409	20.7 ± 24.0
$N_h = 24$	94.0%	1.021 ± 0.383	19.3 ± 15.5
$N_h = 30$	93.0%	1.228 ± 0.644	21.5 ± 23.1

divergence junction exists upstream of Cart B, where it can reroute *before* committing to the congested path. We compare **Mesh ON** vs. **Mesh OFF** using identical seeds, measuring anticipation time ΔT_{ant} (how many seconds earlier Cart B reroutes), backtracking distance, and total makespan.

With mesh enabled, Cart B reroutes 6.82 ± 1.43 s earlier on average, completely eliminating backtracking (0.00 m vs.

TABLE VI: Decoupled Fleet Size Scalability ($N_{humans} = 10$, $N = 100$ Trials).

Fleet Size (N_c)	Success (%)	Makespan (s)	V2V Packets
$N_c = 2$	97.0%	25.67 ± 11.22	5.0 ± 5.0
$N_c = 4$	93.0%	32.31 ± 13.46	32.3 ± 30.8
$N_c = 6$	80.0%	39.69 ± 15.79	55.1 ± 42.0
$N_c = 8$	83.0%	37.32 ± 15.06	74.4 ± 73.6
$N_c = 10$	80.0%	39.68 ± 16.29	107.7 ± 93.6
$N_c = 12$	78.0%	41.25 ± 15.79	134.9 ± 111.4

TABLE VII: Mechanism Experiment A: V2V Mesh Anticipation ($N = 50$ controlled trials).

Condition	ΔT_{ant} (s)	Backtrack (m)	Makespan (s)
Mesh ON	6.82 ± 1.43	0.00 ± 0.00	18.34 ± 2.87
Mesh OFF	0.00 ± 0.00	4.21 ± 1.94	24.96 ± 3.62
Reduction	$+6.82$ s earlier	-4.21 m	-26.5%

4.21 m) and reducing makespan by 26.5% ($p < 0.001$, Welch's t -test). This confirms that the V2V anticipatory mechanism delivers its claimed benefit under the topological conditions it was designed for.

2) *Experiment B: Corridor Mutex Lock ($N = 50$ trials)*: Two carts approach the same single-file corridor from opposite ends. Arrival timing is randomized ($\pm[0, 3]$ s offset). We compare **Lock ON** vs. **Lock OFF** and measure head-on conflict rate, timeout rate, and average corridor traversal time.

With the mutex lock disabled, 22% of trials timeout due to head-on deadlocks and an average of 1.82 ± 0.79 head-on conflicts per trial occur. Lock ON completely eliminates deadlocks (0.00 ± 0.00) and reduces corridor traversal time by 60.8% ($p < 0.001$). This directly validates the R_{lock} mechanism's core contribution.

VII. CONCLUSION & FUTURE RESEARCH DIRECTIONS

This paper introduced the Distributed Dynamic Route Optimization (D²RO) framework powered by Socially-Weighted Distributed Graph Optimization (SW-DGO). By formalizing a 5-component edge traversal cost function $C(u, v, t) = w_D D + w_M W_{mesh} + w_H H_{prox} + w_R R_{lock} + w_S S_{trolley}$, D²RO resolves the fundamental failure modes of prior Multi-Agent Path Finding and reactive navigation paradigms. Extensive Monte Carlo evaluations ($N = 100$ trials per condition) demonstrate a 97.0% mission success rate in the primary supermarket benchmark with 0.00 ± 0.00 corridor deadlocks and 0.59 ± 5.90 intimate personal space violations ($p < 0.001$ vs. all baselines). Performance scales gracefully to 92.0% in clinical hospital and 80.0% in airport terminal environments under increasing architectural complexity. Incremental D* Lite vertex repair times remain within 0.18 ms–1.23 ms across all evaluated crowd densities, well within real-time control loop constraints.

A. Theoretical Guarantees & Deadlock Prevention

- 1) **Heuristic Admissibility & Incremental Optimality**: Because the Euclidean distance heuristic $h(s, s_{goal}) \leq$

TABLE VIII: Mechanism Experiment B: Corridor Mutex Lock ($N = 50$ controlled trials).

Condition	Head-on Conflicts	Timeout Rate	Traversal (s)
Lock ON	0.00 ± 0.00	0.0%	12.47 ± 2.11
Lock OFF	1.82 ± 0.79	22.0%	31.83 ± 9.74

$c^*(s, s_{\text{goal}})$ is strictly admissible and consistent, D* Lite guarantees optimal path extraction with respect to the currently observed edge cost field $C(u, v, t)$ while recomputing only the subgraph perturbed by dynamic obstacles [8].

- 2) **Deadlock Freedom in Single-File Bottlenecks:** In single-file passages where corridor width is insufficient for bidirectional passing, opposing agents entering simultaneously produce reactive live-locks ($F_{\text{net}} = 0$). D²RO prevents deadlocks by assigning directional reservations and broadcasting ‘LOCK_REQUEST’ packets that dynamically inflate the reverse edge cost to $R_{\text{lock}} = \infty$. Opposing carts detect this infinity cost and immediately detour through parallel aisles or hold in Turnout Alcoves (V_{alcove}), empirically eliminating head-on deadlocks across all evaluated trials ($N_{\text{deadlock}} = 0.00 \pm 0.00$).

B. Real-World Physical Considerations & Deployment Constraints

- 1) **RF Signal Attenuation & Steel Fixture Multipath:** Metallic retail shelf gondolas and merchandise packaging attenuate 2.4 GHz RF signals by -15 to -25 dBm. D²RO addresses this through Sub-GHz (868/915 MHz) and dedicated short-range communications (5.9 GHz IEEE 802.11p / DSRC) multi-hop TTL forwarding with autonomous exponential penalty decay ($W_{\text{mesh}}(t) = W_0 \cdot \exp(-\lambda t)$), ensuring that intermittent packet loss does not permanently poison the routing graph.
- 2) **Indoor Positioning & Sensor Fusion:** To mitigate wheel odometry drift on variable flooring (polished supermarket tiles, clinical vinyl, terminal carpets), ground truth localization is maintained via an Extended Kalman Filter (EKF) fusing 100 Hz wheel encoders, 6-DOF IMU gyros, Ultra-Wideband (UWB) transceiver trilateration (Decawave DWM1000), and 2D LiDAR scan-matching against architectural CAD floorplans.
- 3) **Payload Invariance & Non-Holonomic Kinodynamics:** Autonomous carts undergo substantial payload shifts (15 kg empty to 65 kg fully loaded, a +333% mass increase), altering the center of gravity and rotational moment of inertia. D²RO modulates linear acceleration (a_{max}) and steering curvature (κ_{max}) based on real-time motor current draw and load-cell readings to prevent wheel scrubbing or tip-over during turns.

C. Future Research Directions

- 1) **Hybrid Learning-Guided Search:** Future investigations will explore coupling Graph Neural Networks (GNNs) or Deep Reinforcement Learning (e.g., PRIMAL / Learn

to Follow) for global sub-goal allocation with the deterministic D²RO engine for micro-level kinodynamic path execution.

- 2) **Heterogeneous Multi-Agent Ecosystems:** Expanding SW-DGO to mixed-fleet environments comprising autonomous delivery pods, heavy floor-scrubbing AGVs, and human-operated wheelchairs by introducing vehicle-specific agility weights into the R_{lock} reservation tensor.

DATA AND CODE AVAILABILITY

To support scientific reproducibility and open science, all simulation source code, experimental benchmarking harnesses, verification test suites, raw CSV datasets ($N = 100$ Monte Carlo trials across 5 algorithms, 5 cost configurations, and 3 cross-domain architectural environments), and high-resolution vector figures are available under the MIT open-source license at: <https://github.com/polla-fattah/D2RO>.

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